

```
library(ggplot2)

data(diamonds)

head(diamonds)
str(diamonds)
summary(diamonds)
```

[Show hidden output](#)

This is formatted as code

1. How many observations and variables are contained in the dataset?

Answer: The diamonds dataset contains 53,940 observations with 10 variables.

2. Identify the response variable and explain what it measures.

Answer: The response variable is the price, which measures the price of diamond in US dollars.

3. Identify the quantitative explanatory variable used in this assessment and describe its measurement.

Answer: The quantitative explanatory variable is carat, which measures the weight of the diamond.

4. Identify the categorical explanatory variable and list its five categories.

Answer: The categorical explanatory is cut. The five categories includes fair, good, very good, premium, and ideal.

5. Select one category of cut as the reference category and explain why a reference category is necessary when using dummy variables.

```
library(ggplot2)

data(diamonds)
diamonds$cut <- factor(diamonds$cut, ordered = "FALSE")

diamonds$cut <- relevel(diamonds$cut, ref = "Ideal")

contrasts(diamonds$cut)
```

A matrix: 5 × 4 of type dbl

	Fair	Good	Very Good	Premium
Ideal	0	0	0	0
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1

2. Explain what the all-zero category represents.

Answer: The all zero category is Ideal, which is the reference group with value of 0 for all dummy variables that serves as a baseline.

3. Explain why five dummy variables should not be included together with an intercept.

Answer: The five dummy variables including the Ideal, Fair, Good, Very Good, and Premium should not be included with an intercept because one must be left out as a reference group. Since the dummy variables always sum to 1.

```
model1<-lm(price~carat+cut, data=diamonds)
summary(model1)
```

```
Call:
lm(formula = price ~ carat + cut, data = diamonds)

Residuals:
    Min       1Q   Median       3Q      Max
-17540.7  -791.6   -37.6    522.1  12721.4

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2074.55      14.23  -145.77  <2e-16 ***
carat       7871.08      13.98   563.04  <2e-16 ***
cutFair     -1800.92      39.34   -45.77  <2e-16 ***
cutGood      -680.59      24.00   -28.36  <2e-16 ***
cutVery Good -290.79      17.24   -16.87  <2e-16 ***
cutPremium   -361.85      16.69   -21.68  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1511 on 53934 degrees of freedom
Multiple R-squared:  0.8565, Adjusted R-squared:  0.8565
F-statistic: 6.437e+04 on 5 and 53934 DF, p-value: < 2.2e-16
```

1. Write the estimated regression equation from the R output.

Answer: The estimated equation is: Price = - 20074.55 - 7871.08(Carat) - 1800.92(Fair) - 680.59(Good) - 290.79(VeryGoo) - 361.85(Premium)

2. Interpret the coefficient of carat in the context of diamond price.

Answer: The estimated proce of a diamond increases by \$7,871.08.

3. Interpret one dummy-variable coefficient relative to the reference category.

Answer: Compared with the Ideal, a Good cut diamond is predicted to have a \$680.59 lower.

4. Interpret the R² of the model.

Answer: This means that the percentage of 85.65 in idamond price is explained by a carat and cut. While the remaining percentage is due to other factors.

5. Based on the additive model, do the regression lines for the different cut categories have the same slope or different slopes? Explain.

Answer: They have the same slope. As seen in the model, there is no iinteraction between the carat and cut. While the dummy variables only change intercept.

```
model2 <- lm(price ~ carat * cut, data = diamonds)

summary(model2)
```

```
anova(model1, model2)
anova(model2)
```

Call:

```
lm(formula = price ~ carat * cut, data = diamonds)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-14878.3  -793.0   -23.0    546.3  12706.2
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2300.37     19.46  -118.234 < 2e-16 ***
carat          8192.39     23.57   347.568 < 2e-16 ***
cutFair         461.30     86.57    5.329 9.93e-08 ***
cutGood        -122.35     49.35   -2.479 0.013170 *
cutVery Good   -117.29     33.71   -3.479 0.000503 ***
cutPremium     -79.53     32.07   -2.480 0.013155 *
carat:cutFair  -2267.90     76.05  -29.820 < 2e-16 ***
carat:cutGood   -712.76     52.67  -13.533 < 2e-16 ***
carat:cutVery Good -256.42     37.89   -6.768 1.32e-11 ***
carat:cutPremium -384.64     34.18  -11.253 < 2e-16 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1498 on 53930 degrees of freedom

Multiple R-squared: 0.8591, Adjusted R-squared: 0.859

F-statistic: 3.653e+04 on 9 and 53930 DF, p-value: < 2.2e-16

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	53934	123212493961	NA	NA	NA	NA
2	53930	120989369443	4	2223124519	247.7348	2.89282e-211

A anova: 4 × 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
carat	1	729127440120	729127440120	325002.4612	0.00000e+00
cut	4	6133201436	1533300359	683.4558	0.00000e+00
carat:cut	4	2223124519	555781130	247.7348	2.89282e-211
Residuals	53930	120989369443	2243452	NA	NA

1. State the null and alternative hypotheses for the overall carat x cut interaction.

Answer: There is no interaction between the carat and cut and the slope are the same for all cut categories. There is an interaction between carat and cut and at least one slope is different.

2. Using the regression output, identify the interaction coefficients and their corresponding p-values.

Answer: As observed in the model, all the interaction terms have p-values less than 0.05, therefore they are statistically significant to each other.

4. At $\alpha = 0.05$, state your statistical decision and explain what it means in the context of diamond prices.

Answer: We reject the null hypothesis, since the p-value is less than 0.05. This means that the interaction between carat and cut is significant and the effect of carat in pricing differs depending on its cut.

5. Explain the practical difference between the additive model and the interaction model.

Answer: In the additive model, it assumes that carat has the same effect on price for all cut categories. While, the interaction model allowed the effect of carat on price to be different for each category.

```
library(ggplot2)
ggplot(diamonds, aes(x = carat, y = price, color = cut)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_color_manual(values = c("red", "blue", "green", "orange", "purple"))
  labs(
    title = "Diamond Price versus Carat by Cut",
    x = "Carat",
    y = "Price (US Dollars)"
  )
```

